

● RAG Learning

4.2 Congruence, Similarity & Transformations

Pearson Edexcel GCSE Mathematics 1MA1 — Geometry and measures — spec refs G5, G6, G7, G8, G19 — Papers 1 (non-calculator), 2 and 3 (calculator)

This topic covers the four transformations (translation, reflection, rotation, enlargement), how to recognise and prove that two shapes are **congruent** (identical) or **similar** (enlargements of each other), and how to use those facts to find missing lengths. In 1MA1 the Higher tier adds negative scale factors, combinations of transformations and invariance (G8), and the link between lengths, areas and volumes of similar figures.

4.2.1 Congruent Triangles and Simple Proofs (G5, G6)

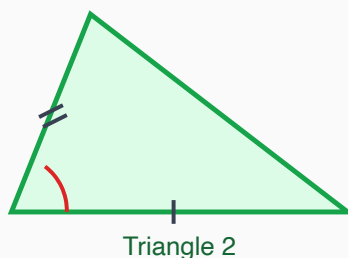
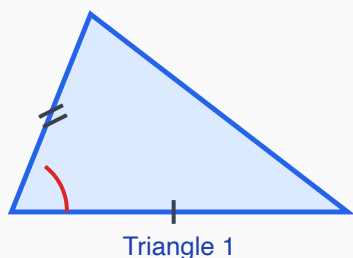
Two shapes are **congruent** when they are identical in shape *and* size: every corresponding side and every corresponding angle is equal. A congruent copy can be produced by translating, rotating or reflecting the original (or any combination of these).

For triangles, Edexcel expects you to know four tests (G5). If any one of them holds, the triangles are congruent:

Test	What you must show is equal	Watch out for
SSS	All three pairs of sides	—
SAS	Two pairs of sides and the angle <i>between</i> them	The angle must be the included angle
ASA (also AAS)	Two pairs of angles and one pair of corresponding sides	The sides must be in matching positions

RHS	A right angle, the hypotenuses and one other pair of sides	Right-angled triangles only
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SAS: two marked sides and the angle between them match $\Rightarrow$ congruent



Exam Tip: Three equal angles (AAA) do **not** prove congruence — they only prove the triangles are *similar*. Two sides with an angle that is not between them (SSA) is also not a valid test, apart from the right-angled case RHS.

Setting out a congruence proof

G6 asks you to use angle facts and congruence to build **simple proofs**. Write one statement per line, give a reason in brackets for each, and finish by naming the test. Then, if the question asks, use "corresponding sides/angles of congruent triangles are equal" to finish.

Worked Example: PQRS is a parallelogram whose diagonals cross at M. Prove that triangles PMQ and RMS are congruent, and hence that the diagonals bisect each other.

Angle MPQ = angle MRS (alternate angles, $PQ \parallel SR$)

Angle MQP = angle MSR (alternate angles, $PQ \parallel SR$)

$PQ = RS$ (opposite sides of a parallelogram are equal)

So triangles PMQ and RMS are **congruent (ASA)**. Corresponding sides are equal, so $PM = MR$ and $QM = MS$: the diagonals bisect each other.

Worked Example: ABC is a triangle with $AB = AC$. Prove that angle ABC = angle ACB (the base angles of an isosceles triangle are equal).

Let M be the midpoint of BC. $AB = AC$ (given), $BM = CM$ (M is the midpoint), AM is common.

So triangles ABM and ACM are congruent (SSS), and therefore angle $ABM = \text{angle } ACM$, i.e. angle $ABC = \text{angle } ACB$. ■

4.2.2 Translations, Reflections and Rotations (G7)

A transformation maps an **object** to an **image**. Edexcel questions either ask you to *carry out* a transformation on a grid or to "**describe fully the single transformation**". In a "describe fully" answer, the type of transformation earns one mark and each required detail earns another.

Transformation	Details needed to describe it fully	Image congruent to object?
Translation	Column vector	Yes
Reflection	Equation of the mirror line	Yes
Rotation	Centre, angle and direction	Yes
Enlargement	Scale factor and centre	No — similar (unless the scale factor is 1 or -1)

Translation

A **translation** moves every point the same distance in the same direction. The column vector $\begin{pmatrix} a \\ b \end{pmatrix}$ means a units right and b units up; negative values mean left and down.

Worked Example: Translate the triangle with vertices $(2, -1)$, $(5, -1)$, $(2, 3)$ by $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$.

Subtract 3 from every x -coordinate and add 4 to every y -coordinate: $(-1, 3)$, $(2, 3)$, $(-1, 7)$.

Reflection

Each image point is the same perpendicular distance from the mirror line as the object point, but on the other side. Points on the mirror line stay where they are.

Mirror line $y = 0$ (x -axis): $(x, y) \rightarrow (x, -y)$ Mirror line $x = 0$ (y -axis): $(x, y) \rightarrow (-x, y)$

Mirror line $y = x$: $(x, y) \rightarrow (y, x)$ Mirror line $y = -x$: $(x, y) \rightarrow (-y, -x)$

Mirror line $x = a$: new $x = 2a - x$ Mirror line $y = b$: new $y = 2b - y$

Worked Example: Reflect $(6, 2)$ in the line $x = 3$, and reflect $(4, 3)$ in the line $y = -1$.

$(6, 2)$ is 3 units right of $x = 3$, so its image is 3 units left: $x = 2(3) - 6 = 0$, giving $(0, 2)$.

$(4, 3)$ is 4 units above $y = -1$, so its image is 4 units below: $y = 2(-1) - 3 = -5$, giving $(4, -5)$.

Rotation

A rotation turns every point through the same angle about a fixed **centre**. Tracing paper may be used in the exam. Tracing paper is allowed in all three 1MA1 papers — ask for it. For rotations about the origin:

90° clockwise: $(x, y) \rightarrow (y, -x)$ 90° anticlockwise: $(x, y) \rightarrow (-y, x)$ 180°:
 $(x, y) \rightarrow (-x, -y)$

Worked Example: Rotate the point $(4, 5)$ through 90° clockwise about $(1, 2)$.

Position relative to the centre: $(4 - 1, 5 - 2) = (3, 3)$.

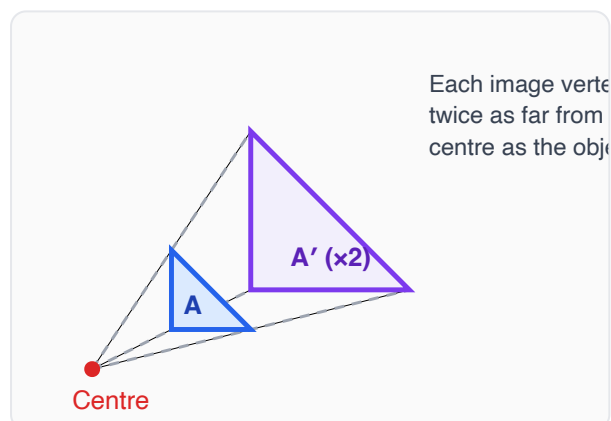
Apply 90° clockwise, $(x, y) \rightarrow (y, -x)$: $(3, -3)$.

Add the centre back on: $(1 + 3, 2 - 3) = (4, -1)$.

Exam Tip: "Describe fully" a rotation needs all three of **centre, angle and direction** (180° needs no direction). Giving two transformations ("reflect then translate") when the question says *single* transformation scores no marks.

4.2.3 Enlargement (G7)

An **enlargement** with scale factor k and centre C multiplies the distance of every point from C by k , along the line through C . Every length is multiplied by k and every angle stays the same, so the image is **similar** to the object.



Scale factor k	What happens
$k > 1$	Image is bigger and further from the centre
$0 < k < 1$ (fractional)	Image is smaller and closer to the centre (both tiers)
$k < 0$ Higher	Image is on the other side of the centre and upside down

$$\text{Scale factor} = \frac{\text{length on image}}{\text{matching length on object}}$$

Image of P : $P' = C + k(P - C)$ (find the vector from the centre, multiply by k , add back to the centre)

Worked Example: Enlarge the points $(7, 4)$ and $(10, 7)$ by scale factor $\frac{1}{3}$, centre $(1, 1)$.

$(7, 4)$: vector from centre $(6, 3)$; one third of it is $(2, 1)$; image $= (1 + 2, 1 + 1) = (3, 2)$.

$(10, 7)$: vector from centre $(9, 6)$; one third is $(3, 2)$; image $= (4, 3)$.

Finding the scale factor and centre

Compare a pair of matching sides to get k . To find the centre on a grid, draw straight lines through each object vertex and its image vertex; they meet at the centre. Without a grid you can use $C = \frac{P' - kP}{1 - k}$.

Worked Example: Triangle A has vertices $(1, 2)$, $(2, 2)$, $(1, 4)$. Triangle B has vertices $(-3, -2)$, $(0, -2)$, $(-3, 4)$. Describe fully the single transformation that maps A onto B.

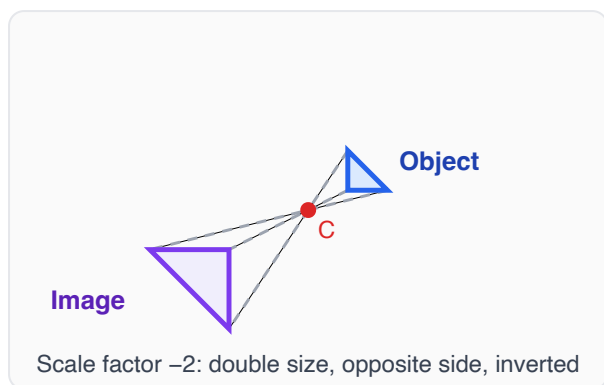
Side $(1, 2)-(2, 2)$ has length 1; the matching side $(-3, -2)-(0, -2)$ has length 3, so $k = 3$.

Centre: $C = \frac{(-3, -2) - 3(1, 2)}{1 - 3} = \frac{(-6, -8)}{-2} = (3, 4)$. Check with $(1, 4)$: $(3, 4) + 3(-2, 0) = (-3, 4) \checkmark$

Enlargement, scale factor 3, centre $(3, 4)$.

Negative scale factors Higher

For $k < 0$, go from the centre to an object point, multiply that distance by $|k|$, and travel that far in the *opposite* direction. The image is inverted — it looks as though it has also been rotated by 180° .



Worked Example: Enlarge the triangle with vertices $(3, 2)$, $(5, 2)$, $(3, 3)$ by scale factor -2 , centre $(1, 1)$.

Vectors from the centre: $(2, 1)$, $(4, 1)$, $(2, 2)$. Multiply by -2 : $(-4, -2)$, $(-8, -2)$, $(-4, -4)$.

Add the centre back: $(-3, -1)$, $(-7, -1)$, $(-3, -3)$.

Exam Tip: Scale factor -1 about a centre has exactly the same effect as a rotation of 180° about that centre, so either description earns the marks. A scale factor between 0 and 1 is still called an *enlargement* in the exam, even though the shape gets smaller.

4.2.4 Combining Transformations and Invariance (G8) Higher

G8 is Higher-only in 1MA1: you must describe the changes and the invariance produced by combinations of rotations, reflections and translations. Track the coordinates of each vertex through each step, then identify the single transformation that takes the original straight to the final image.

Worked Example: Triangle T has vertices $(1, 1)$, $(4, 1)$, $(1, 2)$. T is rotated 90° clockwise about the origin, and the image is then reflected in the y -axis. Describe fully the single transformation that maps T onto the final image.

Rotate, $(x, y) \rightarrow (y, -x)$: $(1, -1)$, $(1, -4)$, $(2, -1)$.

Reflect in the y -axis, $(x, y) \rightarrow (-x, y)$: $(-1, -1)$, $(-1, -4)$, $(-2, -1)$.

Overall $(x, y) \rightarrow (-y, -x)$, so this is a **reflection in the line $y = -x$** . (The point $(1, 1)$ maps to $(-1, -1)$, which confirms it is not on the mirror line.)

A point is **invariant** if it is unchanged by a transformation; a line is invariant if it maps onto itself.

- Reflection: every point on the mirror line is invariant, and so is every line perpendicular to the mirror.
- Rotation (not 360°): only the centre is invariant.
- Translation (non-zero vector): no invariant points, but lines parallel to the vector are invariant.
- Enlargement ($k \neq 1$): only the centre is invariant.

Facts that save time:

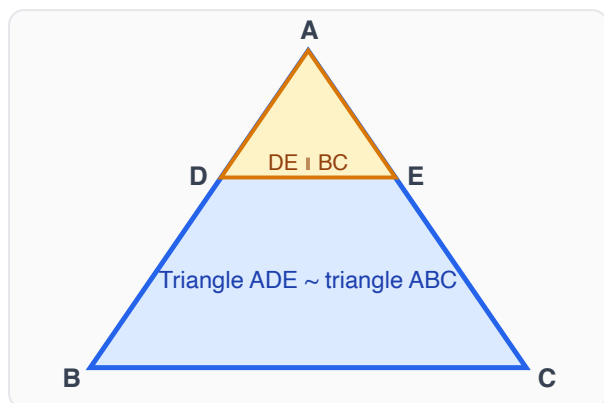
- Reflecting in two **parallel** mirror lines gives a translation through twice the gap between them, e.g. reflect in $x = 4$ then $x = 6$: translation $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$.
- Reflecting in two **intersecting** lines gives a rotation about the crossing point through twice the angle between them.
- Successive translations add: $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ then $\begin{pmatrix} -5 \\ 4 \end{pmatrix}$ is the single translation $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$.
- The order usually matters — check by working through coordinates.

4.2.5 Similar Shapes and Missing Lengths (G6, G7, G19)

Two shapes are **similar** if one is an enlargement of the other: all corresponding **angles are equal** and all corresponding **lengths are in the same ratio**, the scale factor k .

Two triangles are similar if you can show any one of: all angles equal (two pairs are enough); all three pairs of sides in the same ratio; or two pairs of sides in the same ratio with the included

angles equal.



A line drawn parallel to one side of a triangle cuts off a smaller triangle similar to the whole one: the angle at A is shared and the other two pairs are corresponding angles.

Worked Example: In the diagram, $AE = 6$ cm, $EC = 3$ cm and $DE = 8$ cm. Work out BC .

$AC = 6 + 3 = 9$ cm, so the scale factor from ADE to ABC is $9 \div 6 = 1.5$.

$BC = 8 \times 1.5 = \mathbf{12 \text{ cm}}$.

Exam Tip: Find the scale factor from a pair of **whole** corresponding sides (AC with AE), never from the extra piece EC . Redraw the two triangles separately, the same way round, before matching sides.

Worked Example: The lines AE and BD cross at C, and AB is parallel to DE. $AB = 6$ cm, $DE = 15$ cm, $BC = 4$ cm and $CE = 12.5$ cm. Work out CD and AC .

Angle $ACB = \text{angle } ECD$ (vertically opposite) and angle $ABC = \text{angle } EDC$ (alternate), so triangles ABC and EDC are similar with $A \leftrightarrow E$, $B \leftrightarrow D$.

Scale factor from ABC to EDC $= 15 \div 6 = 2.5$.

$CD = 4 \times 2.5 = \mathbf{10 \text{ cm}}$; $AC = 12.5 \div 2.5 = \mathbf{5 \text{ cm}}$.

4.2.6 Areas and Volumes of Similar Figures (G19) Higher

In 1MA1, Foundation G19 covers lengths in similar figures only; the Higher statement adds **areas and volumes**. If corresponding lengths are in the ratio $1 : k$, then corresponding areas are in the ratio $1 : k^2$ and volumes in the ratio $1 : k^3$. "Area" includes surface area and the area of any face or label; "volume" includes capacity, and mass when the objects are made of the same material.

Length scale factor $k \Rightarrow$ area scale factor $k^2 \Rightarrow$ volume scale factor k^3

Working back: $k = \sqrt{\text{area scale factor}} = \sqrt[3]{\text{volume scale factor}}$

Lengths	Areas	Volumes
1 : 4	1 : 16	1 : 64
2 : 5	4 : 25	8 : 125
3 : 4	9 : 16	27 : 64
3 : 5	9 : 25	27 : 125

Worked Example: Two mathematically similar vases are 12 cm and 20 cm tall. The smaller vase holds 324 cm^3 and the larger has surface area 450 cm^2 . Work out the capacity of the larger vase and the surface area of the smaller vase.

$$k = \frac{20}{12} = \frac{5}{3}.$$

$$\text{Capacity of larger} = 324 \times \left(\frac{5}{3}\right)^3 = 324 \times \frac{125}{27} = \mathbf{1500 \text{ cm}^3}.$$

$$\text{Surface area of smaller} = 450 \div \left(\frac{5}{3}\right)^2 = 450 \times \frac{9}{25} = \mathbf{162 \text{ cm}^2}.$$

Worked Example: Two similar statues are made from the same metal. They have masses 2 kg and 16 kg. The smaller statue is 15 cm tall and needs 0.3 litres of paint. Work out the

height of the larger statue and how much paint it needs.

Mass scale factor = $16 \div 2 = 8$ is the volume scale factor, so $k = \sqrt[3]{8} = 2$.

Height = $15 \times 2 = \mathbf{30\text{ cm}}$. Paint covers area, so multiply by $k^2 = 4$: $0.3 \times 4 = \mathbf{1.2\text{ litres}}$.

Exam Tip: Always convert to the **length** scale factor first. Given areas and asked for a volume: square root, then cube. Multiplying a volume by the area scale factor (or vice versa) is the most common error in these questions.

Summary

Idea	Key fact
Congruent triangles (G5)	SSS, SAS, ASA/AAS, RHS — not AAA or SSA
Simple proofs (G6)	Statement + reason on each line, name the test, then use corresponding parts
Translation	Column vector
Reflection	Equation of the mirror line
Rotation	Centre, angle, direction
Enlargement	Scale factor and centre; fractional k gives a smaller image
Negative scale factor Higher	Image on opposite side of the centre, inverted
Combinations and invariance (G8) Higher	Work through coordinates; invariant points map to themselves
Similar figures (G19)	Lengths $\times k$; Higher areas $\times k^2$, volumes $\times k^3$